2.4 GHz FMCW Radar — Horn Antenna Simulation

David Brown

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The horn antenna was simulated in Ansys HFSS (Student 2025 R2). Nothing has been built yet. Plots are in Section 3.

1 Results

Pyramidal horn on WR-340, mouth 263.8 × 193.1 mm, coax probe feed. Band: 2400–2483.5 MHz.

Check	Result	Requirement	
Match (S11), worst in band, original probe (28 / 43.7 mm)	−5.2 dB	≤ −10 dB	**FAIL**
Match (S11), worst in band, final probe (27 / 17.7 mm)	−24.3 dB	≤ −10 dB	**PASS**
Gain at 2.400 / 2.442 / 2.483 GHz	13.24 / 13.24 / 13.27 dBi	12.5–14 dBi	**PASS**
Beamwidth, E / H plane	34.4° / 36.4°	theory 33.4° / 35.2°	**PASS**
TX → RX leakage, horns 290 mm apart	−49.4 dB	≤ −35 dB	**PASS**
Match with probe off by ±1 mm	−14.5 dB	≤ −10 dB	**PASS**
Mesh check (8.8k / 11.9k / 28.8k cells)	0.14 dB gain spread	≤ 0.2 dB	**PASS**
Copper walls vs. perfect conductor	+0.01 dB gain	≤ 0.2 dB	**PASS**

Change made: probe moved from 28 mm long, 43.7 mm from the back wall, to 27 mm long, 17.7 mm from the back wall (picked from a 1 mm sweep of 57 runs).

2 Summary

All checks pass after the probe move (27 mm long, 17.7 mm from the back wall).

3 Figures

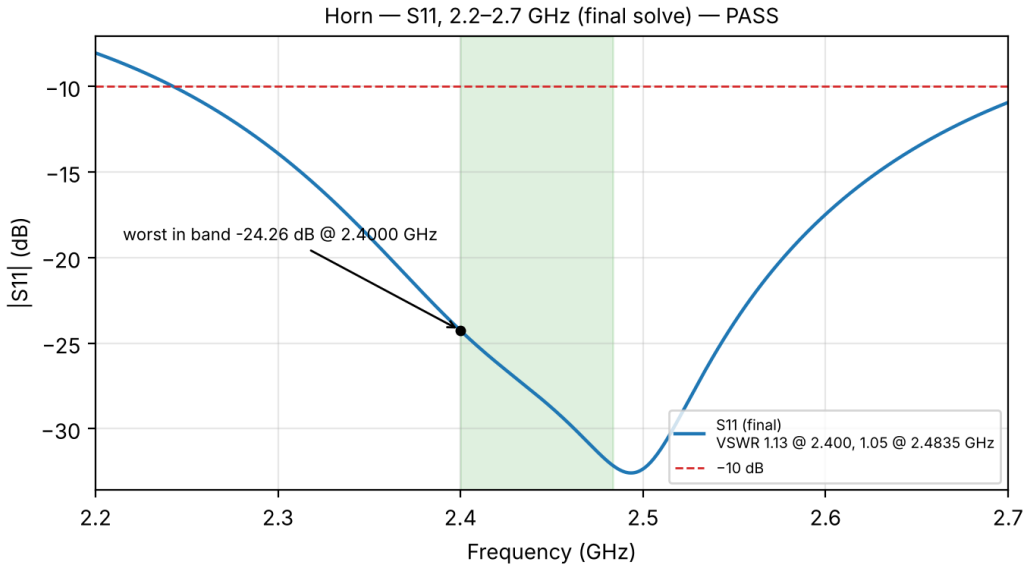


Figure 1: Match (S11) across the band, final probe

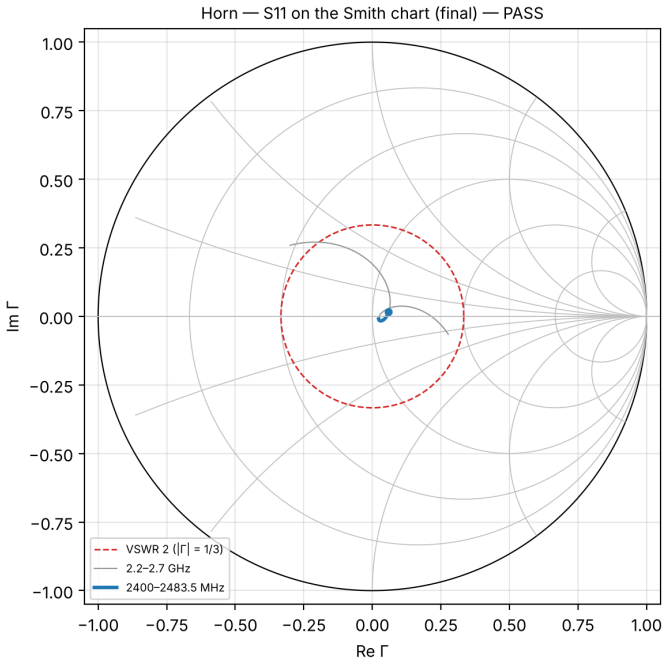


Figure 2: Smith chart, final probe

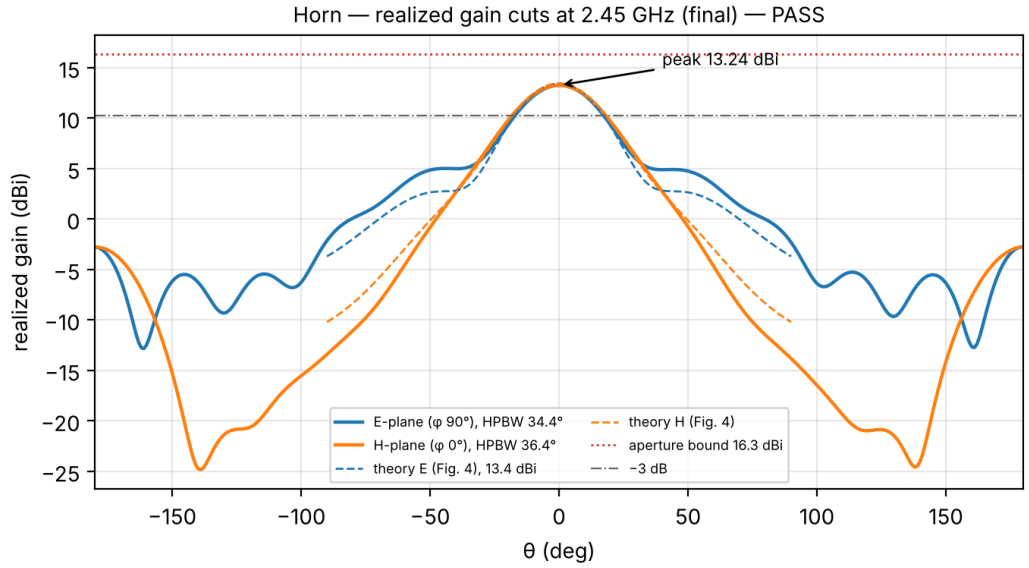


Figure 3: Gain patterns, E and H plane

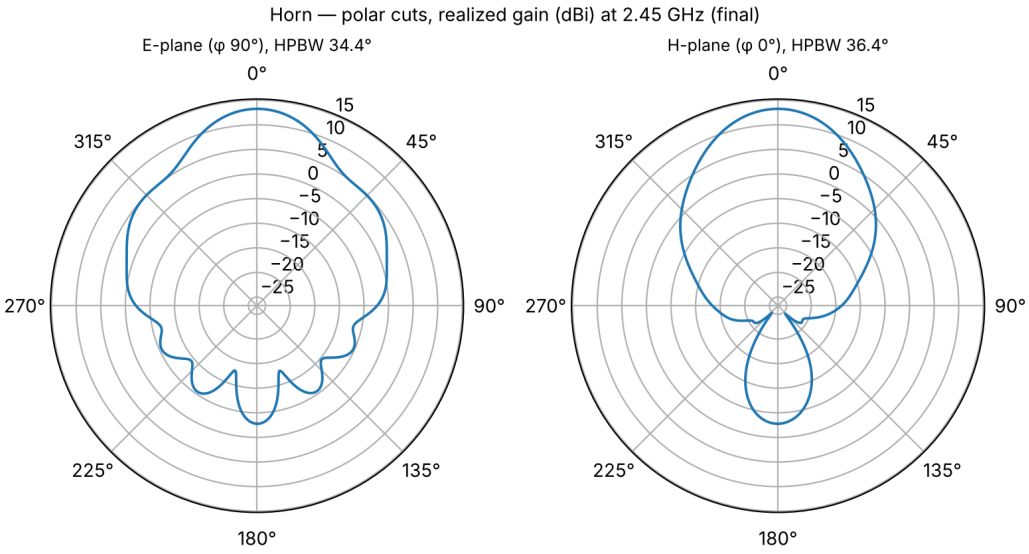


Figure 4: Gain patterns, polar

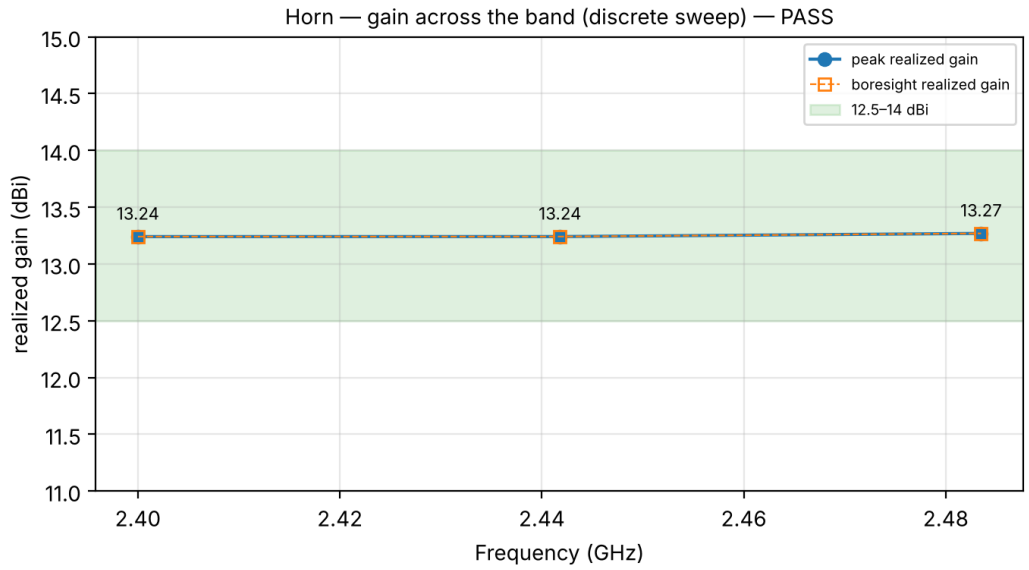


Figure 5: Gain straight ahead across the band

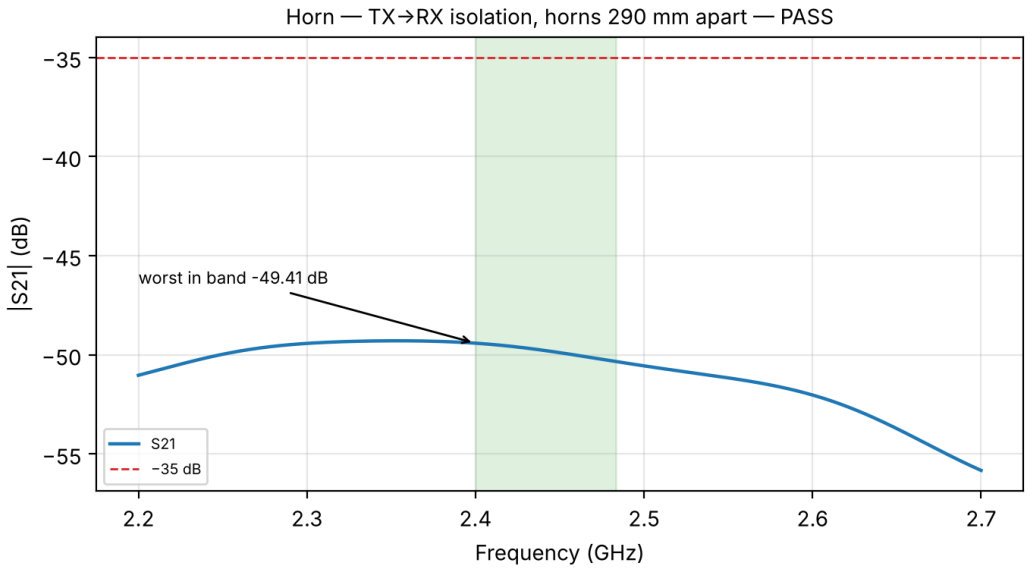


Figure 6: TX to RX leakage, horns 290 mm apart

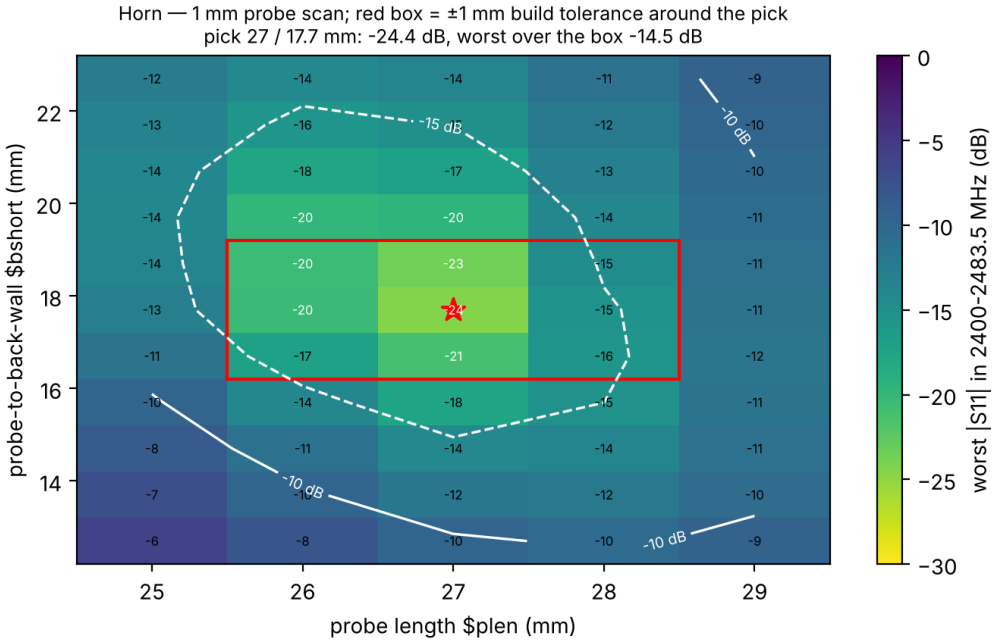


Figure 7: 1 mm probe sweep: worst match in band

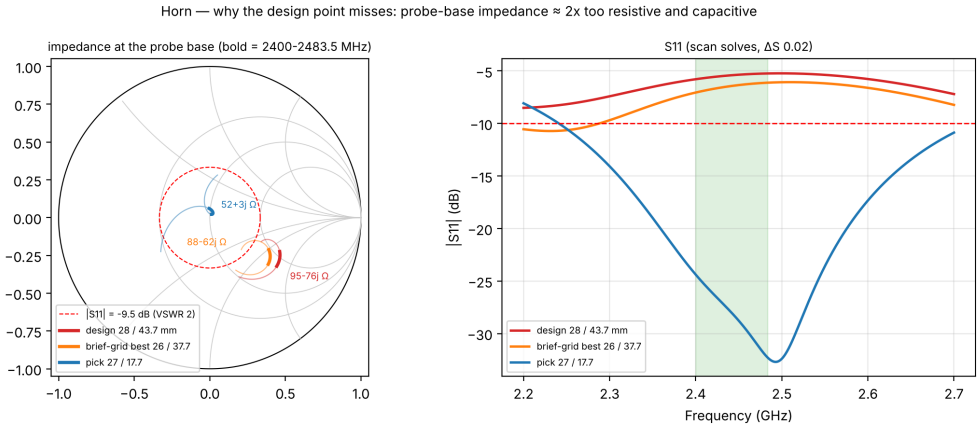


Figure 8: Why the original probe position didn't match

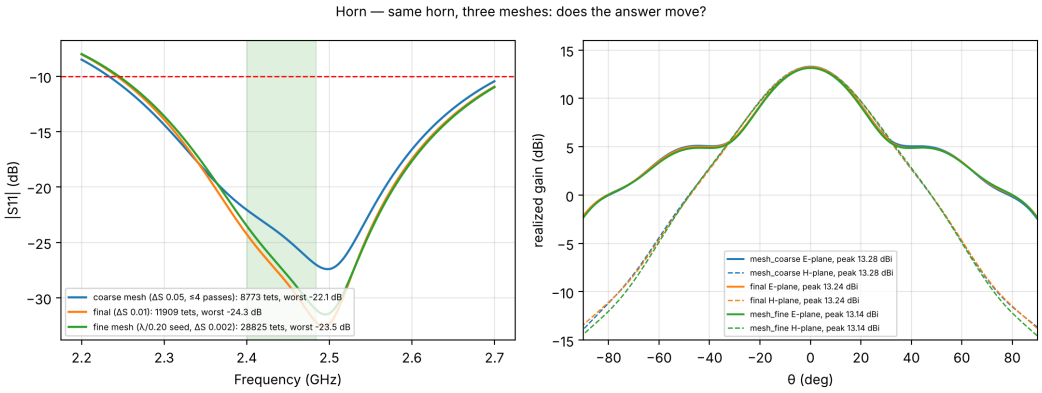


Figure 9: Mesh check

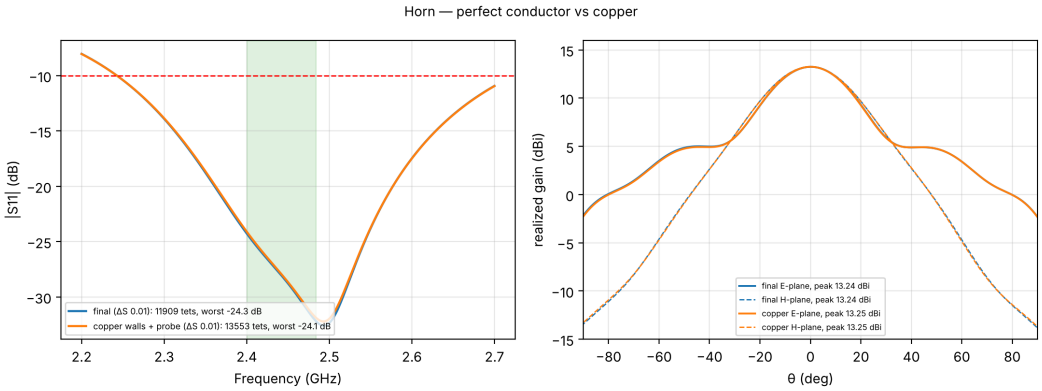


Figure 10: Copper vs. perfect conductor

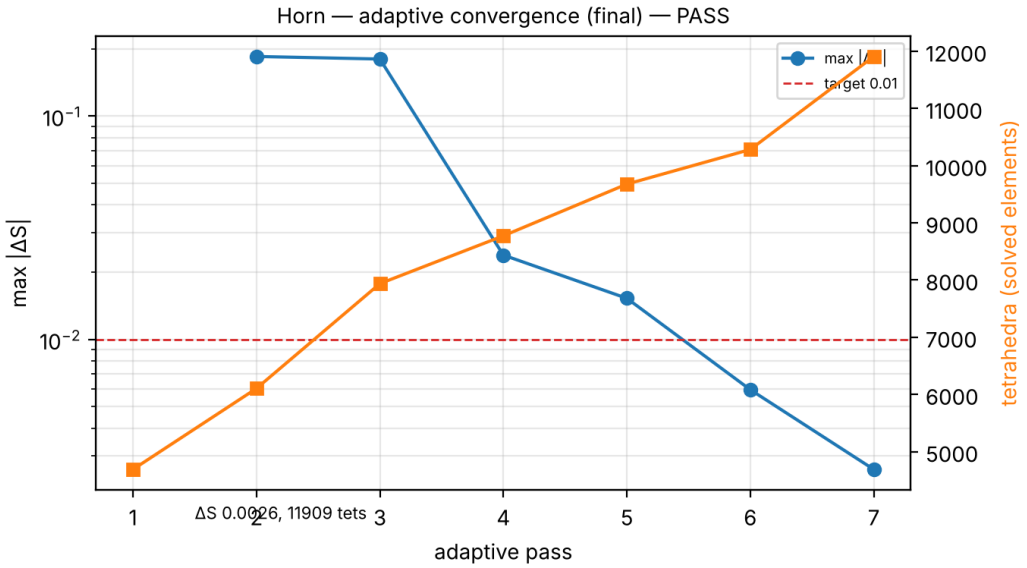


Figure 11: Mesh refinement